15.5.6 Wrap-Up: Cool Down

1. Your friend was absent today and will need the notes from class. Write a quick note to help them understand the difference between the theoretical and experimental probability of a simple experiment.

Unit 15, Lesson 6: Simulating Small and Large Numbers of Trials

Benchmark

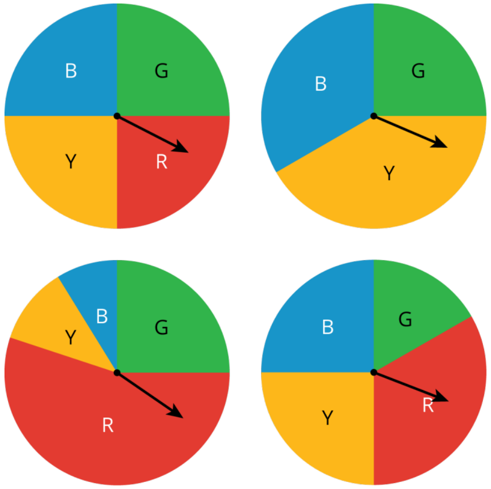
MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Targets

- ☐ I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.
- ☐ I can use results from computer-simulated events to informally define the Law of Large Numbers.

15.6.1 Warm-Up: Which One Doesn't Belong?

1. Four spinners are shown.



a. Which spinner doesn't belong?

b. Explain your reasoning.

15.6.2 Exploration: Comparing Experimental Results to Expected Results

Work with your partner to complete the following.

1. When performing the simple experiment of flipping a coin over repeated trials, imagine each of the resulting situations listed in the table.

   a. Complete the table to describe the results.

Situation	Is the result surprising?		Is the result possible?	
	Yes	**No**	**Yes**	**No**
You flip the coin once, and it lands heads-up.	○	○	○	○
You flip the coin twice, and it lands heads-up both times.	○	○	○	○
You flip the coin 100 times, and it lands heads-up all 100 times.	○	○	○	○

   b. If you flip the coin 100 times, explain how many times you would expect the coin to land heads-up.

   c. With your partner, discuss some other results that would not be surprising if you flip the coin 100 times.

2. A spinner is shown.

a. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Red	Purple	Orange	Blue	Yellow	Green
Theoretical Probability						

b. Sara used a computer-simulated spinner to perform different numbers of trials with the spinner. The results show the percentage of times it landed on each color in each simulation.

Circle the experimental probabilities that are more than 1% different from the theoretical probability.

Color	Red	Purple	Orange	Blue	Yellow	Green
Simulation A (6 spins)	0%	16.7%	16.7%	33.3%	0%	33.3%
Simulation B (60 spins)	16.7%	15%	20%	13.3%	10%	25%
Simulation C (600 spins)	17.7%	19%	18%	15.8%	14.5%	15%
Simulation D (6000 spins)	16.9%	16%	17.1%	16.6%	16.7%	17.1%

c. Discuss with your partner what you notice when comparing the experimental probabilities to the theoretical probabilities.

d. In which simulation(s) were the experimental probabilities the most different from the theoretical probabilities?

e. In which simulation(s) were the experimental probabilities the most similar to the theoretical probabilities?

f. Imagine Sara used the computer-simulated spinner to perform $600{,}000$ spins. Describe how you think the experimental probabilities of landing on each color would compare to the theoretical probabilities.

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results

A ***simulation*** is an approximate imitation of a statistical experiment, often done with a computer program to examine the statistics of a large quantity of trials.

1. Complete the statement.

   As the number of trials in a simulation of a simple experiment increases, the ○ experimental ○ theoretical probability gets ○ closer to ○ farther from the ○ experimental ○ theoretical probability.

2. Imagine flipping a coin 3 times, and it has come up heads once. The experimental probability of landing on heads after 3 trials is $\frac{1}{3}$. Discuss with your partner whether flipping the coin one more time is guaranteed to result in landing on heads so the experimental probability, $\frac{2}{4}$, will equal the theoretical probability $\frac{1}{2}$.

When performing simple experiments, the experimental probabilities may differ from theoretical probabilities due to random variation. However, as the number of repetitions increases, experimental probabilities will typically better approximate the theoretical probabilities.

Simulations are useful when it is hard or time consuming to gather enough information to estimate the probability of some event.

3. Consider the situations described in the table.

Situation	Description
A	The local meteorologist in Ocala, FL says there is a 50% chance of rain tomorrow.
B	A science teacher at Hamilton County High School puts the names of all students in a bag and randomly selects which of 6 groups they will sit in for his new seating chart.
C	The U.S. Centers for Disease Control and Prevention (CDC) released data in 2016 showing that $\frac{1}{3}$ of all adults in the US do not get enough sleep on a regular basis.

   Explain which situation could be simulated using the spinner from the Exploration.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

1. Four situations are described.

Situation	Description
A	In a small lake, 25% of the fish are female. You capture a fish, record whether it is male or female, and toss the fish back into the lake. If you repeat this process 5 times, what is the probability that at least 3 of the 5 fish are female?
B	Elena makes about 80% of her free throws. Based on her past successes with free throws, what is the probability that she will make exactly 4 out of 5 free throws in her next basketball game?
C	On a game show, a contestant must pick one of three doors. In the first round, the winning door has a vacation. In the second round, the winning door has a car. What is the probability of winning a vacation and a car?
D	Your choir is singing in 4 concerts. You and one of your classmates both learned the solo. Before each concert, there is an equal chance the choir director will select you or the other student to sing the solo. What is the probability that you will be selected to sing the solo in exactly 3 of the 4 concerts?

Work with your partner to match each situation to a simulation on the next page.

Simulation	Situation
Toss a standard number cube 2 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 1 or 2 appeared both times to estimate the probability.	
Make a spinner with four equal sections labeled $1, 2, 3,$ and 4. Spin the spinner 5 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 4 appears 3 or more times to estimate the probability.	
Toss a fair coin 4 times and record the outcomes. Repeat this process many times, and find the proportion of the simulations in which exactly 3 heads appear to estimate the probability.	
Place 8 blue chips and 2 red chips in a bag. Shake the bag, select a chip, record its color, and then return the chip to the bag. Repeat the process 4 more times to obtain a simulated outcome. Then repeat this process many times and find the proportion of the simulations in which exactly 4 blues are selected to estimate the probability.	

15.6.5 Wrap-Up: Reflection

1. What is something that is now "squared away" in your mind about simulating the probability of events?

2. What are three "points" that are important about simulating the probability of events?

3. What is something that is still "circling" your brain about simulating the probability of events?

Unit 15, Lesson 7: Experimental and Theoretical Probabilities from Simulations

Benchmarks

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Target

☐ I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.